FOURIER SERIES

Definition of a Periodic Function:

When at equal intervals of abscissa 'x', the value of each ordinate $f(x)$ repeats itself,

i.e$f(x) = f(x+T)$, for all x, then $y = f(x)$ is called a periodic function having period T.

eg. $sin\ x$ and $cos\ x$ are periodic of period 2π,

since $\sin x = \sin(x+2\pi) = \sin(x+4\pi) = \ldots..$

and $\cos x = \cos(x+2\pi) = \cos(x+4\pi) = \ldots..$

Definition:

Let $f(x)$ be a periodic function of period 2π, defined in the interval $(c, c+2\pi)$, satisfying

Dirichlet's Conditions as **(i)** $f(x)$ and its integrals are finite and single valued,

(ii) $f(x)$has discontinuities finite in number ,

(iii) $f(x)$ has finite number of maxima and minima ,

then $f(x)$ can be expanded as an infinite trigonometric series as:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

which is called **Fourier Series**, where $a_0, a_n, b_n (n = 1,2,3, \ldots)$ are called Fourier coefficients or Fourier constants.

EULER'S FORMULAE:

The Fourier series for the function $f(x)$ in the interval $c < x < c+2\pi$ is given by

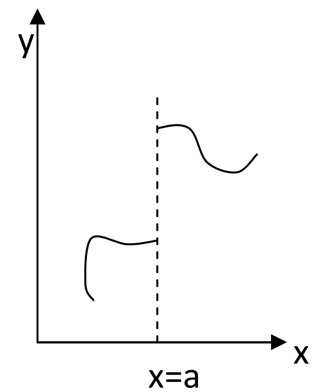
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where, $a_0 = \frac{1}{\pi}\int_c^{c+2\pi} f(x)dx$, $a_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x).\cos nxdx$, $b_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x).\sin nxdx$

These values of a_0, a_n, b_n represented above are known as Euler's Formulae.

FUNCTIONS HAVING POINTS OF DISCONTINUITY:

The function may have a finite number of points of discontinuity, i.e its graph may consists of a finite number of different curves given by different equations. Even then such a function is expressible as a Fourier series.



If in the interval $(c, c+2\pi)$, $f(x)$ is defined by

$$f(x) = f_1(x), \quad c < x < a$$
$$= f_2(x), \quad a < x < c + 2\pi$$

where $x = a$ is the point of discontinuity.

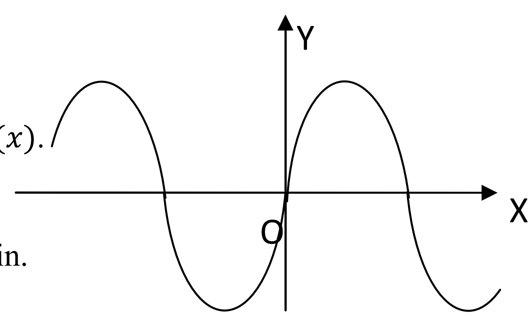
At a point $x = a$, there is a finite jump in the graph of the function.

(See the figure).

Both the limits, limit on the left and the limit on the right exists and are different.

At such a point, Fourier series gives the value of $f(x)$ as the arithmetic mean of these limits. i.e at $x = a,\ f(a) = \frac{1}{2}\left(\lim_{x\to a-} f(x) + \lim_{x\to a+} f(x)\right)$

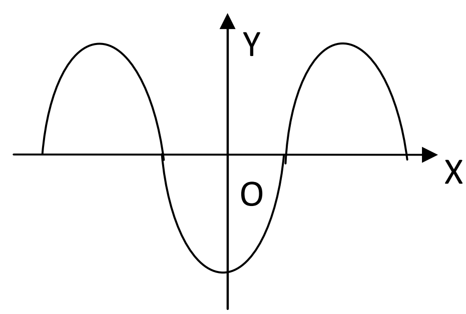
ODD AND EVEN FUNCTIONS:



1. A function $f(x)$ is said to be odd function if $f(-x) = -f(x)$.

e.g $x^3, \sin x, \tan x$ are odd functions.

Graphically, an odd function is symmetrical about the origin.



2. A function $f(x)$ is said to be even function if $f(-x) = f(x)$.

e.g $x^2, \cos x, \sec x$are even functions.

Graphically an even function is symmetrical about the y – axis.

X	**Even**	**Odd**
Even	Even	Odd
Odd	Odd	Even

Note: Odd and even functions can be determined only in the interval of type $(-a, a)$ for any a, where zero must be the mid – point of the interval.

Fourier Series for f(x) [Even / Odd / Neither Even nor Odd] in different interval	
INTERVAL	**FOURIER SERIES**
$(0, 2\pi)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx) + \sum_{n=1}^{\infty} (b_n \sin nx)$ $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx$; $a_n = \frac{1}{\pi}\int_0^{2\pi} f(x).\cos nxdx$; $b_n = \frac{1}{\pi}\int_0^{2\pi} f(x).\sin nxdx$
$(0, 2l)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{l}\right)\right) + \sum_{n=1}^{\infty} \left(b_n \sin\left(\frac{n\pi x}{l}\right)\right)$ $a_0 = \frac{1}{l}\int_0^{2l} f(x)dx$; $a_n = \frac{1}{l}\int_0^{2l} f(x).\cos\left(\frac{n\pi x}{l}\right) dx$; $b_n = \frac{1}{l}\int_0^{2l} f(x).\sin\left(\frac{n\pi x}{l}\right) dx$
$(-\pi, \pi)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx) + \sum_{n=1}^{\infty} (b_n \sin nx)$ $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$; $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x).\cos nx\, dx$; $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x).\sin nx\, dx$
$(-l, l)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{l}\right)\right) + \sum_{n=1}^{\infty} \left(b_n \sin\left(\frac{n\pi x}{l}\right)\right)$ $a_0 = \frac{1}{l}\int_{-l}^{l} f(x)dx$; $a_n = \frac{1}{l}\int_{-l}^{l} f(x).\cos\left(\frac{n\pi x}{l}\right) dx$; $b_n = \frac{1}{l}\int_{-l}^{l} f(x).\sin\left(\frac{n\pi x}{l}\right) dx$
Even function in $(-\pi, \pi)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx)$ $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$; $a_n = \frac{2}{\pi}\int_0^{\pi} f(x).\cos nx\, dx$; $b_n = 0$
Even function in $(-l, l)$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{l}\right)\right)$ $a_0 = \frac{2}{l}\int_0^{l} f(x)dx$; $a_n = \frac{2}{l}\int_0^{l} f(x).\cos\left(\frac{n\pi x}{l}\right) dx$; $b_n = 0$
Odd function in $(-\pi, \pi)$	$f(x) = \sum_{n=1}^{\infty} (b_n \sin nx)$ $a_0 = 0$; $a_n = 0$; $b_n = \frac{2}{\pi}\int_0^{\pi} f(x).\sin nx\, dx$
Odd function in $(-l, l)$	$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin\left(\frac{n\pi x}{l}\right)\right)$ $a_0 = 0$; $a_n = 0$; $b_n = \frac{2}{l}\int_0^{l} f(x).\sin\left(\frac{n\pi x}{l}\right) dx$

PARSEVAL'S IDENTITY:

If $f(x) = \frac{a_0}{2} + \sum\limits_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{c} + b_n \sin\frac{n\pi x}{c}\right)$ is the Fourier series in $(0, 2c)$

then prove that $\frac{1}{c}\int_0^{2c}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right\}$

Intervals	Parseval's identity
$(0, 2c)$	$\frac{1}{c}\int_0^{2c}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2 + b_n^2)\right\}$
$(0, 2\pi)$	$\frac{1}{\pi}\int_0^{2\pi}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2 + b_n^2)\right\}$
$(-c, c)$	$\frac{1}{c}\int_{-c}^{c}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2 + b_n^2)\right\}$
$(-\pi, \pi)$	$\frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2 + b_n^2)\right\}$
half – range cosine series in $(0, c)$	$\frac{2}{c}\int_0^{c}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2)\right\}$
half – range cosine series in $(0, \pi)$	$\frac{2}{\pi}\int_0^{\pi}[f(x)]^2 dx = \left\{\frac{a_0^2}{2} + \sum\limits_{n=1}^{\infty} (a_n^2)\right\}$
half – range sine series in $(0, c)$	$\frac{2}{c}\int_0^{c}[f(x)]^2 dx = \left\{\sum\limits_{n=1}^{\infty} (b_n^2)\right\}$
half – range sine series in $(0, \pi)$	$\frac{2}{\pi}\int_0^{\pi}[f(x)]^2 dx = \left\{\sum\limits_{n=1}^{\infty} (b_n^2)\right\}$

EXERCISE

1. If $f(x) = 2x$, $0 \leq x \leq 2\pi$ find the fourier series for $f(x)$ and find b_{10} and a_4
2. Obtain Fourier series for $f(x) = xcosx$ in $(0, 2\pi)$.
3. Obtain Fourier series for $f(x) = \frac{1}{2}(\pi - x)$ in $(0, 2\pi)$. Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$
4. For $0 < x \leq 2\pi$and $k \neq 0$, prove that $\pi e^{kx} = (e^{2k\pi} - 1).\left(\frac{1}{2k} + \sum_{n=1}^{\infty} \frac{k\cos nx - n\sin nx}{n^2+k^2}\right)$
5. Obtain Fourier series for $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ 0, & \pi \leq x \leq 2\pi \end{cases}$

   Hence, deduce that $\frac{1}{2} = \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \cdots$ or $1 = \sum_{n=1}^{\infty} \frac{2}{(2n-1)(2n+1)}$
6. Obtain Fourier series for $f(x) = \begin{cases} -\pi, & 0 \leq x \leq \pi \\ x - \pi, & \pi \leq x \leq 2\pi \end{cases}$ state the value of the series at $x = \pi$ and

   hence, show that $\frac{\pi^2}{8} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$
7. Obtain the Fourier expansion of $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in the interval $0 \leq x \leq 2\pi$ and $f(x + 2\pi) = f(x)$

   Also deduce that

   (i) $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$ **(ii)** $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$

   (iii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$ **(iv)** $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots$
8. Obtain Fourier series for $f(x) = x^2$ in $0 < x < a$, hence find the sum of the series

   $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$
9. If $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2 - x), & 1 \leq x \leq 2 \end{cases}$. Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$
10. Find Fourier series to represent $f(x) = 4 - x^2$ in the interval $(0,2)$

    Hence, deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$
11. Obtain Fourier series for $f(x) = e^x \; in \; (-\pi, \pi)$. Hence derives the series for $\frac{\pi}{\sin h\pi}$
12. Find the Fourier series for $f(x)$, if $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$

    Hence deduce that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = \frac{\pi^2}{8}$

13. Find the Fourier Series for $f(x) = \begin{cases} \cos x, & -\pi < x < 0 \\ \sin x, & 0 < x < \pi \end{cases}$

14. Find the Fourier expansion of $f(x) = x^2, -\pi \le x \le \pi$, and hence, prove that

(i) $\frac{\pi^2}{6} = \sum_1^\infty \frac{1}{n^2}$, **(ii)** $\frac{\pi^2}{12} = \sum_1^\infty \frac{(-1)^{n+1}}{n^2}$, **(iii)** $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$

(iv) By using Parseval's identity prove that $\sum \frac{1}{n^2} = \frac{\pi^4}{90}$

15. Obtain Fourier series for $f(x)$ in the range $(-\pi, \pi)$, where $f(x) = |x|$

16. Obtain a Fourier series for $f(x) = |\sin x|$ in the range $-\pi < x < \pi$

17. Obtain Fourier series for $f(x) = x \cos x \; in \; (-\pi, \pi)$

18. Obtain Fourier series for $f(x)$ given by $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi < x < 0 \\ 1 - \frac{2x}{\pi}, & 0 < x < \pi \end{cases}$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$

19. Find the Fourier series for $f(x) = x + x^2$ in $-\pi \le x \le \pi$. Hence deduce that :

(i) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$. **(ii)** $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} \ldots = \frac{\pi^2}{8}$.

20. If $f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi \le x \le 0 \\ \frac{\pi}{2} - x, & 0 \le x \le \pi \end{cases}$ Prove that: $f(x) = \frac{4}{\pi}\left(\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots.\right)$

Hence deduce that: $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$ Also deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots$

21. Find the Fourier expansion for $f(x) = x - x^2, -1 < x < 1$

22. Obtain the Fourier expansion for $f(x) = \begin{cases} 0, & -2 < x < -1 \\ 1 + x, & -1 < x < 0 \\ 1 - x, & 0 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$ in the range $(-2, 2)$

23. Find the Fourier series for $f(x) = 1 - x^2$ when $-1 \le x \le 1$.

24. Find the Fourier expansion of $f(x) = \begin{cases} 2, & -2 < x < 0 \\ x, & 0 < x < 2 \end{cases}$

25. Obtain half – range Fourier cosine series for $f(x) = x$ in $0 < x < 2$. Hence deduce that

(i) $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots = \frac{\pi^4}{96}$ **(ii)** $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots = \frac{\pi^4}{90}$

26. Find half – range sine series for $f(x) = x \sin x$ in $(0, \pi)$ and deduce that

(i) $\frac{1}{1^2.3^2} - \frac{3}{5^2.7^2} + \frac{5}{9^2.11^2} - \frac{7}{13^2.15^2} + \cdots = \frac{\pi^2}{64\sqrt{2}}$

(ii) $\frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \frac{1}{11^2} + \cdots = \frac{\pi^2}{8\sqrt{2}}$

27. In $(0, \pi)$ show that

$$x^2 = \frac{2}{\pi}\left\{\left(\frac{\pi^2}{1} - \frac{4}{1^3}\right)\sin x - \frac{\pi^2}{2}\sin 2x + \left(\frac{\pi^2}{3} - \frac{4}{3^3}\right)\sin 3x - \frac{\pi^2}{4}\sin 4x + \left(\frac{\pi^2}{5} - \frac{4}{5^3}\right)\sin 5x - \cdots\right\}$$

28. Obtain a half range cosine series for $f(x) = \begin{cases} kx, & for\ 0 \le x \le \frac{l}{2} \\ k(l-x) & for\ \frac{l}{2} \le x \le l \end{cases}$

Hence deduce the sum $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$

29. Obtain the half – range cosine series for $f(x) = \pi x - x^2$ in $(0, \pi)$ and hence deduce that

(i) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}$

(ii) $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots$

(iii) $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + .. = \frac{\pi^4}{90}$

(iv) $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \cdots = \frac{\pi^4}{96}$

30. Show that, in $0 < x < \pi$, $\cos x = \frac{8}{\pi}\sum_{m=1}^{\infty} \left(\frac{m}{4m^2-1}\right)\sin 2mx$

31. Obtain the half – range **(i)** Cosine Series **(ii)** sine series for $f(x) = lx - x^2$ in $(0, l)$.

Hence deduce that

(i) $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots = \frac{\pi^3}{32}$

(ii) $\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \frac{1}{7^6} + .. = \frac{\pi^6}{960}$

(iii) $\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \frac{1}{4^6} + \cdots = \frac{\pi^6}{945}$

32. Obtain a half – range cosine series for $f(x)$, where $f(x) = \begin{cases} x & 0 < x < \pi/2 \\ \pi - x & \pi/2 < x < \pi \end{cases}$

33. Obtain a half – range sine series in $(0, \pi)$ for $f(x) = x(\pi - x)$ and hence, find the value of $\sum \frac{(-1)^{n+1}}{(2n-1)^3}$

ANSWERS

1. $2x = 2\pi - 4\sum_{n=1}^{\infty}\frac{\sin nx}{n}, a_4 = 0, b_{10} = -0.4$

2. $x\cos x = \pi\cos x - \frac{1}{2}\sin x - 2\sum_{n=2}^{\infty}\frac{n\sin nx}{n^2-1}$

3. $f(x) = \sum_{n=1}^{\infty}\frac{\sin nx}{n}$ Then put $x = \frac{\pi}{2}$

5. $f(x) = \frac{1}{\pi} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{2}{4n^2-1}\cos 2nx + \frac{1}{2}\sin x$

6. $f(x) = -\frac{\pi}{4} + \frac{2}{\pi}\left[\frac{1}{1^2}\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right] + \left[-3\sin x - \frac{1}{2}\sin 2x - \sin 3x + \cdots\right]$

7. $f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{1}{n^2}\cos nx = \frac{\pi^2}{12} + \frac{1}{1^2}\cos x + \frac{1}{2^2}\cos 2x + \frac{1}{3^2}\cos 3x + \cdots$

8. $f(x) = \frac{a^2}{3} + \frac{a^2}{\pi^2}\left[\frac{1}{1^2}\cos\frac{2\pi x}{a} + \frac{1}{2^2}\cos\frac{4\pi x}{a} + \cdots\right] - \frac{a^2}{\pi}\left[\frac{1}{1}\sin\frac{2\pi x}{a} + \frac{1}{2}\sin\frac{4\pi x}{a} + \cdots\right]$

9. $f(x) = \frac{\pi}{2} - \frac{4}{\pi^2}\left[\frac{\cos\pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \cdots\right] = \frac{\pi}{2} - \frac{4}{\pi^2}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}\cos(2n+1)\,\pi x$

10. $f(x) = \frac{8}{3} - \frac{4}{\pi^2}\left[\frac{1}{1^2}\cos\pi x + \frac{1}{2^2}\cos 2\pi x + \cdots\right] + \frac{4}{\pi}\left[\frac{1}{1}\sin\pi x + \frac{1}{2}\sin 2\pi x + \cdots\right]$

11. $f(x) = \frac{2\sin h\pi}{\pi}\left[\frac{1}{2} + \sum_{n=1}^{\infty}\frac{(-1)^n}{(1+n^2)}\cos nx - \sum_{n=1}^{\infty}\frac{n(-1)^n}{(1+n^2)}\sin nx\right]$

12. $f(x) = -\frac{\pi}{4} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n-1}{n^2}\cos nx + \sum_{n=1}^{\infty}\frac{1-2(-1)^n}{n}\sin nx$

13. $f(x) = \frac{1}{\pi} + \frac{1}{2}\cos x - \frac{2}{\pi}\left[\frac{\cos 2x}{3} + \frac{\cos 4x}{15} + \cdots\right] + \frac{1}{2}\sin x - \frac{2}{\pi}\left[\frac{2\sin 2x}{3} + \frac{4\sin 4x}{15} + \cdots\right]$

$= \frac{1}{\pi} + \frac{1}{2}(\cos x + \sin x) + \frac{2}{\pi}\left[\sum_{n=1}^{\infty}\frac{1}{(1-4n^2)}\cos 2nx + \sum_{n=1}^{\infty}\frac{2n}{(1-4n^2)}\sin 2nx\right]$

14. $f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$

15. $f(x) = \frac{\pi}{2} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos(2n-1)\,x$

16. $f(x) = \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos 2n\pi}{4n^2-1}$

17. $f(x) = -\frac{1}{2}\sin x + 2\sum_{n=2}^{\infty}(-1)^n\cdot\frac{n}{n^2-1}\sin nx$

18. $f(x) = \frac{8}{\pi^2}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$

19. $f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx + 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\sin nx$

21. $f(x) = -\frac{1}{3} - \frac{4}{\pi^2}\sum \frac{(-1)^n}{n^2}\cos n\pi x - \frac{2}{\pi}\sum \frac{(-1)^n}{n^2}\sin n\pi x$

22. $f(x) = \frac{1}{4} + \frac{4}{\pi^2}\sum \frac{1}{n^2}\left(1 - \cos\frac{n\pi}{2}\right)\cos\frac{n\pi x}{2}$

23. $f(x) = \frac{2}{3} - \frac{4}{\pi^2}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos n\pi x$

24. $f(x) = \frac{3}{2} - \frac{4}{\pi^2}\left[\frac{1}{1^2}\cos\frac{\pi x}{2} + \frac{1}{3^2}\cos\frac{3\pi x}{2} + \frac{1}{5^2}\cos\frac{5\pi x}{2} + \cdots\right] - \frac{2}{\pi}\left[\frac{1}{1}\sin\frac{\pi x}{2} + \frac{1}{2}\sin\frac{2\pi x}{2} + \frac{1}{3}\sin\frac{3\pi x}{2} + \cdots\right]$

25. $f(x) = 1 - \frac{8}{\pi^2}\left[\frac{1}{1^2}\cos\frac{\pi x}{2} + \frac{1}{3^2}\cos\frac{3\pi x}{2} + \cdots\right]$

26. $f(x) = \frac{\pi}{2}\sin x + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right)\sin 2x + \left(\frac{1}{5^2} - \frac{1}{3^2}\right)\sin 4x + \left(\frac{1}{7^2} - \frac{1}{5^2}\right)\sin 6x + \cdots\right]$

28. $f(x) = \frac{kl}{4} - \frac{8kl}{\pi^2}\left[\frac{1}{2^2}\cos\frac{2\pi x}{l} + \frac{1}{6^2}\cos\frac{6\pi x}{l} + \frac{1}{10^2}\cos\frac{10\pi x}{l} + \cdots\right]$

29. $f(x) = \frac{\pi^2}{6} - \left[\frac{1}{1^2}\cos 2x + \frac{1}{2^2}\cos 4x + \frac{1}{3^2}\cos 6x + \cdots\right]$

31. **(i)** $f(x) = lx - x^2 = \frac{l^2}{6} - \frac{4l^2}{\pi^2}\left[\frac{1}{2^2}\cos\frac{2\pi x}{l} + \frac{1}{4^2}\cos\frac{4\pi x}{l} + \frac{1}{6^2}\cos\frac{6\pi x}{l} + \cdots\right]$

(ii) $f(x) = lx - x^2 = \frac{8l^2}{\pi^3}\left[\frac{1}{1^3}\sin\frac{\pi x}{l} + \frac{1}{3^3}\sin\frac{3\pi x}{l} + \frac{1}{5^3}\sin\frac{5\pi x}{l} + \cdots\right]$

32. $f(x) = \frac{\pi}{4} - \frac{8}{\pi}\left[\frac{1}{2^2}\cos 2x + \frac{1}{6^2}\cos 6x + \frac{1}{10^2}\cos 10x + \cdots\right]$

33. $f(x) = \frac{8}{\pi}\left[\frac{1}{1^3}\sin \pi x + \frac{1}{3^3}\sin 3\pi x + \frac{1}{5^3}\sin 5\pi x + \cdots\right]$